Potential

Building Models to Describe our World

Defining, Calculating Electric Potential, and Real World Examples

Outline

Defining Electric Potential

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Recall: Work-Energy Theorem

If net work is performed on a particle in going from 1 to 2, that work is the change in kinetic energy:

$$W^{net} = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F}^{net} \cdot d\vec{r} = K_2 - K_1 = \Delta K = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

Remember, this was the whole reason for defining kinetic energy. We defined kinetic energy because the difference in it is the net work done.

For conservative forces, define potential energy

- Calculating work can be complicated, as we have to evaluate the integral over the 3-dimensional path over which the force was applied.
- However, if the work does not depend on the path taken, we can define a scalar function of position, $U(\vec{r})$, “potential energy”, that we evaluate at the end points of the path to calculate the work done by a specific force:

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r} = -\left[U(\vec{r}_2) - U(\vec{r}_1)\right] = -\Delta U$$

- Why the minus sign?

For conservative forces, define potential energy

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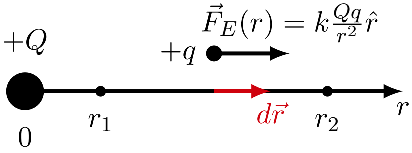
- Why the minus sign? Conservation of energy!

$$W^{net} = \Delta K = -\Delta U \quad \rightarrow \quad \Delta K + \Delta U = 0$$

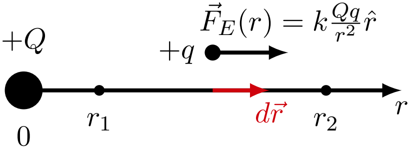
Defining a function for electrical potential energy

- As a test case, use a “fixed” charge $+Q$ at the origin, and calculate the work done by the electric force on a positive charge $+q$ in going from: $\vec{r}_1$ to $\vec{r}_2$
- Analogy with gravity: Use a fixed mass, M, at the origin, and calculate the work done by gravity on a mass, m, in going from position $\vec{r}_1$ to position $\vec{r}_2$
- Choose a path that is always parallel to electric force.



Determining the work done on charge q in going from $\vec{r}_1$ to $\vec{r}_2$

Defining a function for electrical potential energy



Determining the work done on charge q in going from $\vec{r}_1$ to $\vec{r}_2$

Calculate the work done on charge q in going from $\vec{r}_1$ to $\vec{r}_2$:

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F}\cdot d\vec{r} = \int_{\vec{r}_1}^{\vec{r}_2} q\vec{E}\cdot d\vec{r} = kQq\int_{r_1}^{r_2}\frac{1}{r^2}dr$$

$$= kQq\left[\frac{1}{r_1} - \frac{1}{r_2}\right] = -\Delta U = U_1 - U_2$$

Define the potential energy function as (by inspection):

$$\boxed{U(r) = k\frac{Qq}{r} + C}$$

Choosing the constant, C

$$U(r) = k\frac{Qq}{r} + C$$

A simple and convenienent choice is $C = 0$. This implies that the potential energy of q, when it is infinitely far away from Q, is zero. It makes sense, but is ulimately meaningless, since **only differences in potential energy are meaningful**.

$$\boxed{U(r) = k\frac{Qq}{r}}$$

What if q is negative?

Repeat the same exercise if the charge q is negative:

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r} = \int_{\vec{r}_1}^{\vec{r}_2} q\vec{E} \cdot d\vec{r} = q \int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r} = kQq \left[\frac{1}{r_1} - \frac{1}{r_2} \right] = -\Delta U = U_1 - U_2$$

Math is the same since q factors out! That's the whole point of defining the electric field!

$$\boxed{U(r) = k\frac{Qq}{r} + C}$$

The product qQ will take into account that the force changed direction (and thus the sign of the work being done depends on whether qQ is positive or negative).

Electric potential, V

- We defined **electric field**,$\vec{E}$, so that we could easily calculate the force on a charge q of any sign. Electric field is *force per unit charge*:

$$\vec{F} = q\vec{E}$$

- We do the same thing for potential energy! We define **electric potential**, V, as the *potential energy per unit charge*:

$$U = qV$$

For a point charge, we have:

$$\vec{F} = k\frac{qQ}{r^2}\hat{r} \rightarrow \vec{E} = \frac{\vec{F}}{q} = k\frac{Q}{r^2}\hat{r}$$

and:

$$U = k\frac{Qq}{r} \rightarrow V = \frac{U}{q} = k\frac{Q}{r}$$

Properties of electric potential

- Just like potential energy, only difference in potentials are meaningful:

$$\Delta U = q\Delta V$$
$$\Delta V = \frac{\Delta U}{q}$$

- Electric potential (like potential energy) is a **scalar function of position**.
- Potential energy relates to a specific charge, whereas potential relates to the potential energy that any charge would have.
- **Voltage is a difference in potentials** (ΔV is a voltage). Saying that “something is at voltage of $200\,\mathrm{V}$” necessarily implies *relative* to something else (e.g. ground).

Electric potential, V

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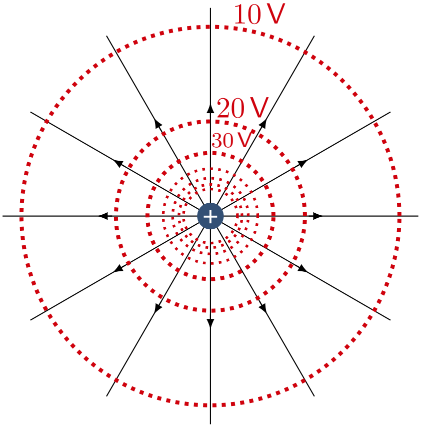
and:

$$U = k\frac{Qq}{r} \rightarrow V = \frac{U}{q} = k\frac{Q}{r}$$

Electric potential

- Only differences in potential are meaningful, and they correspond to differences in potential energy
- All charges, if left free, will try to decrease their potential energy
- For a positive charge, potential energy increases with increasing potential
- For a negative charge, potential energy decreases with increasing potential



Point charge: electric field and potential.

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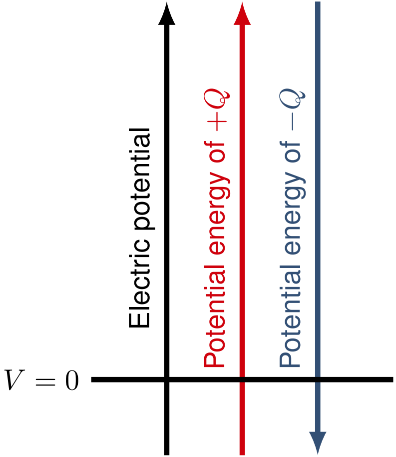
Potential and potential energy.

Electric potential

It doesn't matter where you set $V = 0$ to, as long as you're consistent!



Potential and potential energy.

Potential and Potential Energy

Gravitational Force

- Force:

$$\vec{F} = \frac{GMm}{r^2}\hat{r}$$

- Field (force per unit mass) near **point mass** M:

$$\vec{g} = \frac{GM}{r^2}\hat{r}$$

- Potential energy near **point mass** M:

$$U(r) = -\frac{GMm}{r} + C = -\int \vec{F} \cdot d\vec{r}$$

- "Potential" **from point mass** M (potential energy per unit mass):

$$V(r) = -\frac{GM}{r} + C = -\int \vec{g} \cdot d\vec{r}$$

Electric Force

- Force:

$$\vec{F} = \frac{kQq}{r^2}\hat{r}$$

- Field (force per unit *charge*) near **point *charge*** Q:

$$\vec{E} = \frac{kQ}{r^2}\hat{r}$$

- Potential energy near **point *charge*** Q:

$$U(r) = -\frac{kQq}{r} + C = -\int \vec{F} \cdot d\vec{r}$$

- "Potential" **from point *charge*** Q (potential energy per unit *charge*):

$$V(r) = -\frac{kQ}{r} + C = -\int \vec{E} \cdot d\vec{r}$$

Calculating Electric Potential

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Calculating electric potential

Work done by the force gives potential energy:

$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r} = \int_{\vec{r}_1}^{\vec{r}_2} q\vec{E} \cdot d\vec{r} = q \int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r} = -\Delta U$$

Doing this per charge, gives electric potential

$$-\Delta V = \frac{-\Delta U}{q} = \int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r}$$

Electric potential is to electrical potential energy what electric field is to electric force:

$$\Delta U = -\int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}$$

$$\Delta V = -\int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r}$$

Two methods to determine electric potential

1. **Use the electric field** and integrate it (if you already know the electric field):

$$\Delta V = (V_2 - V_1) = -\int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r}$$

$$V(\vec{r}) = -\int \vec{E} \cdot d\vec{r}$$

2. **Model as a collection of point charges** and sum the potentials (Make sure they share the location of $0\,\mathrm{V}$!):

$$dV = \frac{kdq}{r}$$

$$\therefore V = \int dV = k\int \frac{dq}{r}$$

Using the E-field

This is what we did to determine the potential near a point charge Q

Summing the potentials

This is analogous to how we calculate the electric field for a distributed charge, but much easier since potential is a scalar!

Electric potential near a point charge, Q

- We can calculate the potential energy function from the anti-derivative:

$$V(r) = -\int \vec{E} \cdot d\vec{r} = -\int k\frac{Q}{r^2}dr = k\frac{Q}{r} + C$$

- **If we define zero potential energy at infinity ($C = 0$)**, then this also corresponds to zero electric potential, and we can define electric potential from a point charge as:

$$V(r) = k\frac{Q}{r}$$

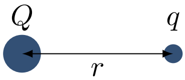
Electric potential energy near point charge, Q

- Found the potential energy of q a distance r from Q:

$$U(r) = \frac{kQq}{r} + C$$

- Convenient choice is $C = 0$
- This means that we explicitly define $U(r) = 0$ when $r \to \infty$:

$$\boxed{U(r) = \frac{kQq}{r}}$$



Point charges separated by distance, r.

Potential from a ring of charge, along symmetry axis

- Break it up into many small point charges, dq.
- Write the potential from a dq:

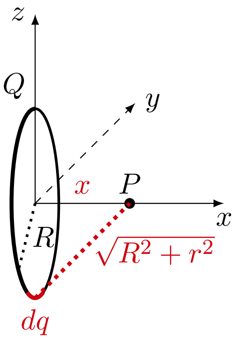
$$dV = \frac{kdq}{r} = \frac{kdq}{\sqrt{R^2 + x^2}}$$

- Sum the potentials (integrate):

$$\therefore V = \int dV = \int_0^Q \frac{kdq}{\sqrt{R^2+x^2}} = \frac{kQ}{\sqrt{R^2+x^2}}$$

Careful!

This only works if all the potentials, dV, have the same locations defined for $0\,\mathrm{V}$ (namely, at infinity, since we chose $C = 0$).



A point charge, dq, on a ring of radius, R.

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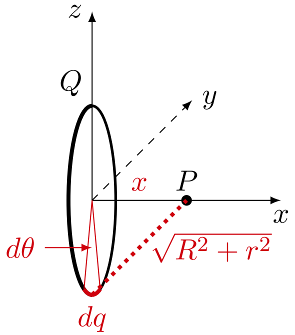


A point charge, dq, on a ring of radius, R.

Easy integrand!

We don't need to express dq with a position variable (e.g. $dq = \lambda R d\theta$) since all quantities in the integrand are constants! But you should try to convince yourself you'll get the same answer!

Potential from a disk carrying surface charge, σ

- Break up the disk into small rings of charge, dq, and radius, r. Each ring has an area dA:

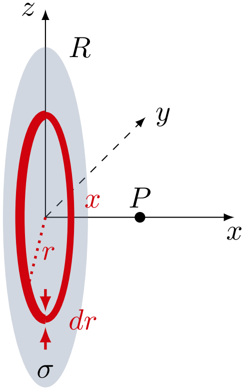
$$dq = \sigma dA = \sigma 2\pi r dr$$

- Write the potential from dq (using our previous result):

$$dV = \frac{kdq}{\sqrt{r^2 + x^2}} = \frac{k\sigma 2\pi r dr}{\sqrt{r^2 + x^2}}$$

- Sum the potentials:

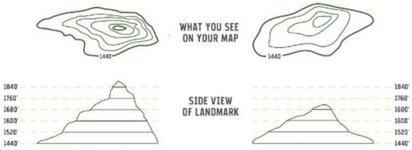
$$\therefore V = \int dV = 2\pi k\sigma \int_0^R \frac{r}{\sqrt{r^2 + x^2}} dr$$



A ring of charge dq on a disk carrying surface charge σ. Think of unfolding the ring into a rectangle of length $2\pi r$ and height dr to determine its area, dA.

What about $\vec{E}$ from V?

- If we know the electric field vector, by using a scalar product ($q\vec{E} \cdot d\vec{r}$), we can find the electric potential (vector $\rightarrow$ scalar)
- How do we go the other way (scalar $\rightarrow$ vector)?
  - Based on the direction in which electric potential increases, we can **determine the direction of the field**.
  - Based on how quickly the potential changes, we can determine the **magnitude of the field**.



Contour lines on a map

Contour lines

Lines (contours) of constant electric potential correspond to lines of constant electric potential energy. Just like lines of constant altitude correspond to lines of constant gravitational potential energy.

$\vec{E}$ from V in 1D

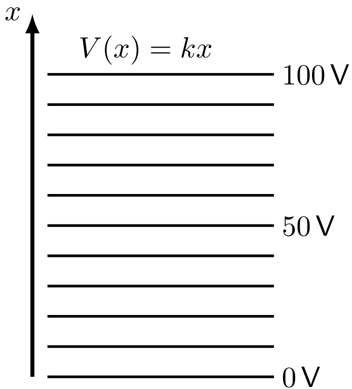
- V from $\vec{E}$:

$$V(x) = -\int E dx$$

- $\vec{E}$ from V

$$\therefore E(x) = -\frac{dV}{dx} = -k$$

- Electric potential is the (negative) anti-derivative of electric field (in 1D).
- Electric field is this the (negative) derivative of electric potential.



Example of a linear potential.

$\vec{E}$ from V in 3D

- V from $\vec{E}$:

$$V(\vec{r}) = V(x, y, z) = -\int \vec{E} \cdot d\vec{r}$$

- $\vec{E}$ from V

$$E_x = -\frac{\partial V}{\partial x}$$
$$E_y = -\frac{\partial V}{\partial y}$$
$$E_z = -\frac{\partial V}{\partial z}$$

- In 3D, we need to know how the potential changes (its derivatives) in each direction. Those (partial) derivatives are the 3 components of the electric field!

Different Notation

The electric field is the (negative) gradient of the electric potential:

$$\begin{aligned}\vec{E} &= -\nabla V \\ &= -\left(\frac{\partial V}{\partial x}\hat{x} + \frac{\partial V}{\partial y}\hat{y} + \frac{\partial V}{\partial z}\hat{z}\right)\end{aligned}$$

Application

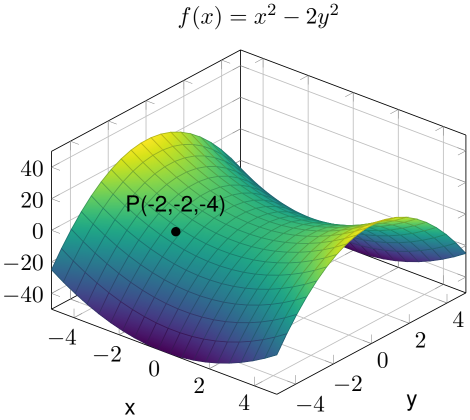
Building Models to Describe our World

Defining, Calculating Electric Potential, and Real World Examples

Recall the gradient vector

- From the partial derivatives, we know how the function changes along the x and y directions
- The gradient tells us the direction in which the function changes the fastest (in which direction the slope is the steepest)
- It is a vector whose magnitude tells us how fast the function is changing in the direction of maximal change.

$$\nabla f(x,y) = \frac{\partial f}{\partial x}\hat{x} + \frac{\partial f}{\partial y}\hat{y}$$
$$= 2x\hat{x} - 4y\hat{y} = (-4, 8)$$



A point, P, is on the surface given by $f(x,y) = x^2 - 2y^2$.

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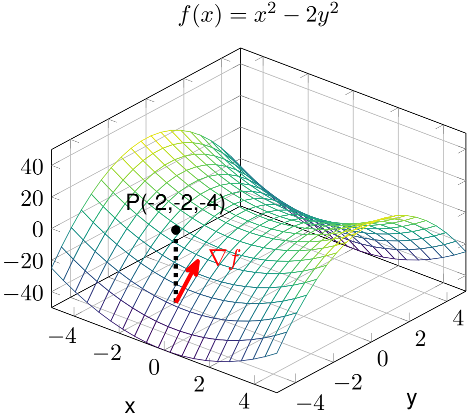


A vector parallel to the gradient vector is projected in the xy-plane. It points in the direction of most rapid increase.

Gradient in physics

- The **gradient of the potential energy** tells us in which direction the **corresponding force** is exerted. The force is exerted in the opposite direction of the gradient vector (the force is exerted in the direction opposite to that which potential energy increases).
  - Think of a contour map in the context of gravity. The contours are regions of constant elevation (namely, constant gravitational potential energy).
- The **gradient of the electric potential** tells us in which direction the **corresponding electric field** is. The field is in the opposite direction of the gradient vector.

Summary: Electric Potential

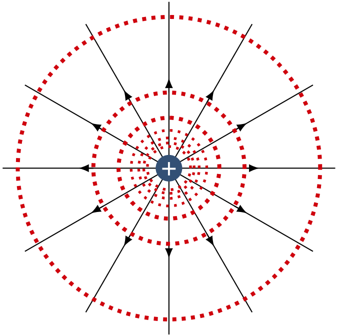
- We started with Coulomb's Law for the **electric force vector** between 2 points charges: $\vec{F}$
- We defined the **electric field vector**, which is independent of the "test" charge, q: $\vec{F} = q\vec{E}$
- Knowing that the force is conservative, we defined a (scalar) **potential energy function**: $U(\vec{r})$
- We defined the **electric potential function**, which is independent of the test charge, q: $U(\vec{r}) = qV(\vec{r})$

$\rightarrow$ We can now describe the motion of electric charges using scalar quantities (electric potential and electric potential energy).

$\rightarrow$ Instead of an electric force vector acting in certain direction, you can think of charges moving to regions of space where they will lower the electric potential energy.

Equipotential surfaces

- Electric potential is a scalar function of position, $V(x, y, z)$.
- Electric potential gives the electric potential energy of a charge at a given position.
- **Equipotential surfaces** are regions of space where electric potential is constant.
- The electric field does not do work along an equipotential surface.
- Conductors are always equipotentials (the net field inside a conductor must be zero in electrostatics, so the potential cannot change inside a conductor).



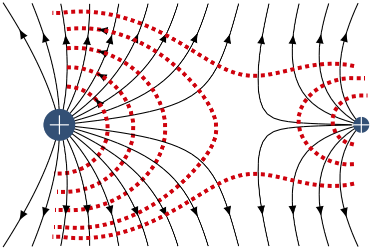
For a point charge, equipotential surfaces consist in concentric spheres; the electric field foes no work along one of these surfaces.

Equipotential surfaces

- Since they are regions of constant electric potential energy, the electric field does no work when moving along an equipotential
- The equipotential surfaces are always perpendicular to the electric field lines:

$$W = \vec{F} \cdot \vec{d} = Fd\cos\theta$$

→ Electric field is always perpendicular to surface of the conductor



Electric field lines and corresponding equipotentials from two unequal positive charges.

Charges on the surface of a conductor

- Charges spread uniformly on the outer **surface** of a conducting sphere.
- Gauss' Law:

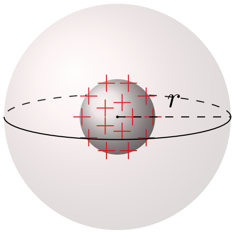
$$(4\pi r^2)E(r) = \frac{Q}{\epsilon_0} \rightarrow E(r) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

- At some distance, r, the potential is given by:

$$V(r) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r} \quad (V(r\rightarrow\infty) = 0)$$

$\rightarrow$ For a charged sphere, evaluating V and E at $r = R$, we thus have:

$$\boxed{V = ER}$$



A charged metallic sphere with gaussian surface

Charges on the surface of a conductor

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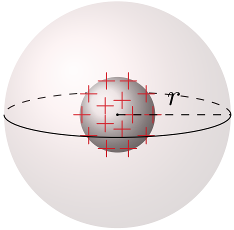


A charged metallic sphere with gaussian surface

Charge density on the sphere

Given the potential on the surface, we can determine the amount of charge (or surface charge density):

$$Q = 4\pi\epsilon_0 RV$$

$$\therefore \sigma = \frac{Q}{4\pi R^2} = \frac{\epsilon_0 V}{R}$$

Connecting two spheres

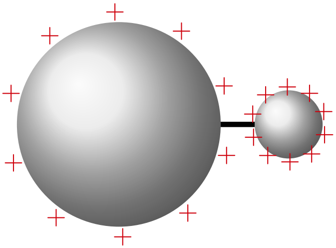
- The two conducting spheres form an equipotential:

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q_1}{R_1} = \frac{1}{4\pi\epsilon_0}\frac{Q_2}{R_2}$$
$$\therefore \frac{Q_1}{R_1} = \frac{Q_2}{R_2}$$

$\rightarrow$ Comparing the surface charge densities:

$$\frac{\sigma_1 4\pi R_1^2}{R_1} = \frac{\sigma_2 4\pi R_2^2}{R_2}$$
$$\therefore \sigma_1 R_1 = \sigma_2 R_2$$

For any charged conducting object, the **charge density will be highest where the radius of curvature is smallest**.



Two connected charged conducting spheres.

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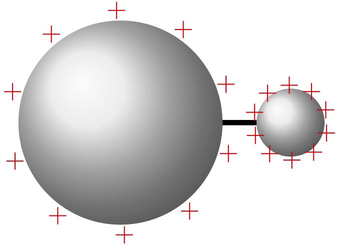


Two connected charged conducting spheres.

Electric field

The potential is the same on each sphere:

$$V = E_1 R_1 = E_2 R_2$$

The **electric field is strongest where radius of curvature is smallest**.

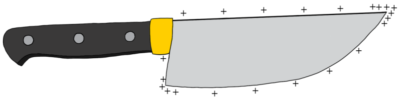
The lightening rod

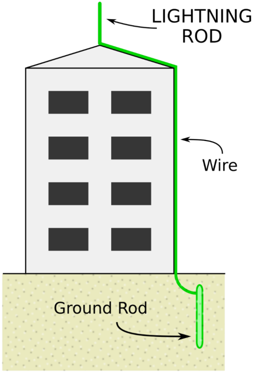
- In general, charges get concentrated at the sharp part of a conductor (the charge density and electric field are highest where the radius of curvature is smallest)
- This means the electric field there can be very high on a sharp point

→ Ideally, the strong electric field allows for corona discharge, and charges can leak from the lighting rod into the cloud

→ If the electric field gets too strong, then the air breaks down, and you get lightning!

So lighting rods do draw lighting, but ideally, they prevent it all together!



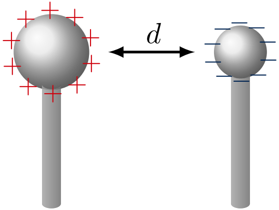


Measuring the Van der Graaf terminal voltage

- We can assume that the electric field is roughly constant between the two spheres (like parallel plates):

$$\Delta V = Ed$$

- For breakdown in air, $E = 3 \times 10^6$ V/m.
- Can measure d on the Van der Graaf to determine its voltage!



The two terminals of a Van der Graaf generator, separated by a distance d.

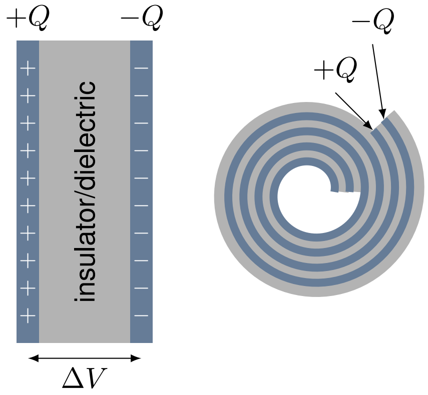
Capacitors

- Capacitors are a way to store charges.
- Two electrodes, always equal and opposite charge, $+Q$ and $-Q$, held at a potential difference, ΔV (imagine parallel plates).
- Geometry determines how much charge you can store on a capacitor for a given voltage, the "capacitance", C:

$$Q = C\Delta V$$

- The capacitor stores electric potential energy:

$$U = \frac{1}{2}\frac{Q^2}{C^2} = \frac{1}{2}C\Delta V^2 = \frac{1}{2}Q\Delta V$$



Different capacitor geometries.

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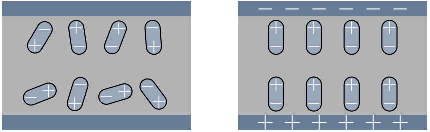

Dielectric materials contain little dipoles; by re-orienting them, we can store energy.

Dielectrics

The capacitance also depends on the properties of the material between the plates. More energy is stored by using a dielectric material (one that can polarize). The dielectric lowers the field between the plates, allowing for more charges at a given voltage.